Act as an operations analyst extracting a process map from a video source. Use the standard process-analysis terms defined in the reading provided (for example: step, resource, run time, capacity, inventory/buffer) to label things consistently. Using only information explicitly stated or shown in the video — not the reading — list every step in this process in chronological order. For each step give: (1) step name, (2) actor/resource, (3) inputs, (4) outputs, (5) any quantitative figures explicitly stated, with a timestamp, (6) whether the step is sequential or parallel to a named step. If a figure is implied but not explicitly stated, write “not stated” — do not estimate

Based on the sources provided, here is the operational process map for handcrafting shuttlecocks at Dodo Shuttles, using standard process-analysis terminology.

Handmade Shuttlecock Production Process Map

Step 1: Feather Washing

- **Actor/Resource:** Israel Mullick and co-workers (Labor); soapy water and buckets (Resources).
- **Inputs:** White duck feathers (Raw Material).
- **Outputs:** Cleaned feathers (WIP).
- **Quantitative Figures:** Scrubbing takes "several seconds" per handful; up to an hour to complete a batch of this size [Segment 1].
- **Sequential or Parallel:** Sequential; the initial step in the process.

Step 2: Initial Rinsing

- **Actor/Resource:** Workers (Labor); water (Resource).
- **Inputs:** Washed feathers.
- **Outputs:** Rinsed feathers.
- **Quantitative Figures:** Not stated.
- **Sequential or Parallel:** Sequential to Step 1.

Step 3: Tinting & Second Rinse

- **Actor/Resource:** Israel Mullick (Labor); fresh water and natural indigo dye (Resources).
- **Inputs:** Rinsed feathers.
- **Outputs:** Tinted feathers.

- **Quantitative Figures:** Not stated.
- **Sequential or Parallel:** Sequential to Step 2; performed "swiftly in tandem" with rinsing and wringing.

Step 4: Wringing

- **Actor/Resource:** Workers (Labor).
- **Inputs:** Tinted/wet feathers.
- **Outputs:** Damp feathers.
- **Quantitative Figures:** Not stated.
- **Sequential or Parallel:** Sequential to Step 3.

Step 5: Drying

- **Actor/Resource:** The sun (Resource).
- **Inputs:** Damp feathers.
- **Outputs:** Clean and dry feathers.
- **Quantitative Figures:** Process takes "three to four hours" [Segment 2].
- **Sequential or Parallel:** Sequential to Step 4.

Step 6: Sorting and Inspection

- **Actor/Resource:** Worker/Co-owner Subo Nyogi (Labor).
- **Inputs:** Dry feathers.
- **Outputs:** Uniform, high-strength feathers (standardized WIP).
- **Quantitative Figures:** Not stated.
- **Sequential or Parallel:** Sequential to Step 5.

Step 7: Trimming (Length)

- **Actor/Resource:** Worker (Labor); shears bolted together (Resource).
- **Inputs:** Inspected feathers.
- **Outputs:** Feathers trimmed to uniform length.
- **Quantitative Figures:** Feathers are trimmed to "exactly three inches long" [Segment 2].
- **Sequential or Parallel:** Sequential to Step 6.

Step 8: Shaping (Edges)

- **Actor/Resource:** Team of workers/cutters (Labor).
- **Inputs:** Trimmed feathers.
- **Outputs:** Shaped feathers with one curved and one straight edge.
- **Quantitative Figures:** Takes "just a few seconds" per feather; workers churn through "hundreds... a day" (Capacity) [Segment 2].
- **Sequential or Parallel:** Sequential to Step 7.

Step 9: Cork Drilling

- **Actor/Resource:** Worker (Labor); drilling machine (the only machine used in the factory).
- **Inputs:** Cork bases imported from China (Raw Material).
- **Outputs:** Drilled cork bases.
- **Quantitative Figures:** Worker drills "16 holes into each base" [Segment 3].
- **Sequential or Parallel:** Narratively sequential to Step 8, but operates as a parallel preparation of the base resource before assembly.

Step 10: Gluing (Assembly)

- **Actor/Resource:** Worker (Labor); glue (Resource).
- **Inputs:** 16 shaped feathers and one drilled cork base.
- **Outputs:** Assembled shuttlecock frame.
- **Quantitative Figures:** "16 [feathers]" are used per shuttlecock [Segment 1]; uses a "quick dab of glue" [Segment 3].
- **Sequential or Parallel:** Sequential to Step 9.

Step 11: Binding

- **Actor/Resource:** Worker (Labor); thread (Resource).
- **Inputs:** Assembled shuttlecock frame.
- **Outputs:** Shuttlecock with a bound "skirt".
- **Quantitative Figures:** Worker makes "two separate rings of thread" [Segment 3].
- **Sequential or Parallel:** Sequential to Step 10.

Step 12: Epoxy Application

- **Actor/Resource:** Worker (Labor); paint brush and epoxy (Resources).
- **Inputs:** Bound shuttlecock.
- **Outputs:** Reinforced shuttlecock.
- **Quantitative Figures:** Applies a "thin coat of epoxy" [Segment 3].
- **Sequential or Parallel:** Sequential to Step 11.

Step 13: Labeling

- **Actor/Resource:** Not stated (Labor); labels (Resource).
- **Inputs:** Finished shuttlecock.
- **Outputs:** Branded shuttlecock.
- **Quantitative Figures:** Not stated.
- **Sequential or Parallel:** Sequential to Step 12.

Step 14: Weighing

- **Actor/Resource:** Israel Mullick (Labor); scale (Resource).
- **Inputs:** Branded shuttlecock.
- **Outputs:** Verified-weight shuttlecock.
- **Quantitative Figures:** Each shuttlecock must weigh "70 grams" [Segment 3].
- **Sequential or Parallel:** Sequential to Step 13.

Step 15: Packaging

- **Actor/Resource:** Israel Mullick (Labor); tubes (Resource/Inventory).
- **Inputs:** Verified shuttlecocks.
- **Outputs:** Packaged tubes for sale.
- **Quantitative Figures:** Stacks "10 of them in a tube"; tubes sell for "200 to 300 rupees" (about 4) [Segments 3, 4].
- **Sequential or Parallel:** Sequential to Step 14.

Process Overview Metrics

- **Total Throughput Time:** From start to finish, a single shuttlecock takes "just 30 minutes to make".
- **Standard Batching:** Production is often "done in stages" (Batch processing).

- **Resource Constraints:** Foreign competition and synthetic alternatives are noted as external factors affecting the factory's capacity and labor pool.